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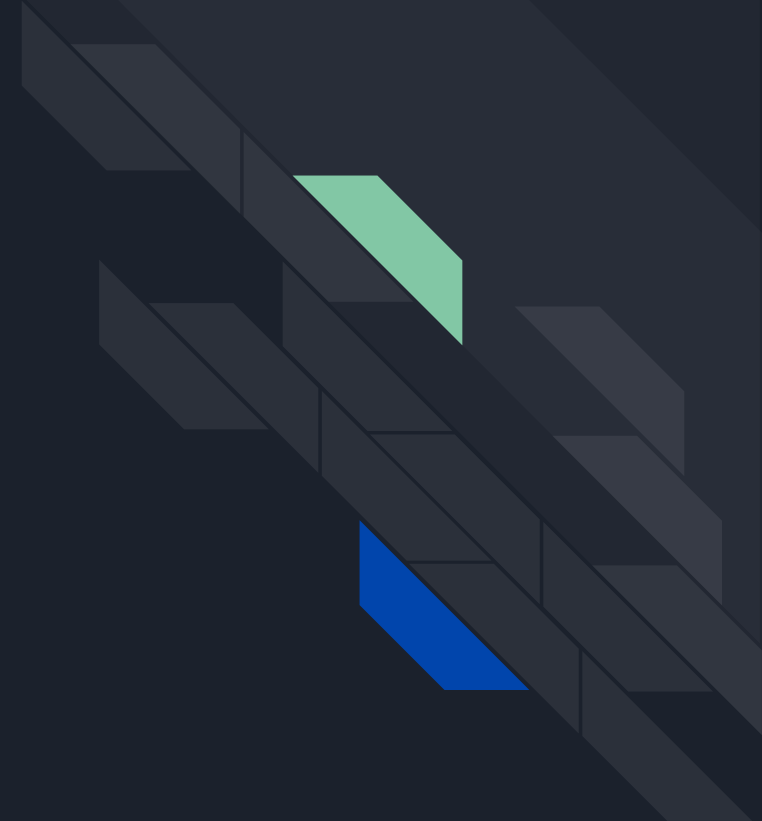
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Introduction

1. Olist is Brazilian e-commerce network that connect vendors and customers across multiple product categories



Project Objectives

1. Provide a convenient platform for customers to discover a wide range of products
2. Provide a smooth shopping experience



Core Business Insights

- Customer segmentation
- Popular product categories
- High demand products
- Preferred payment methods
- Seller performance

Core Business Insights

1. Which customer has the highest total price of purchases?
 - From the result, customers' total purchase costs can range from 13664.08 to 3256.14 units.

SQL QUERY 1

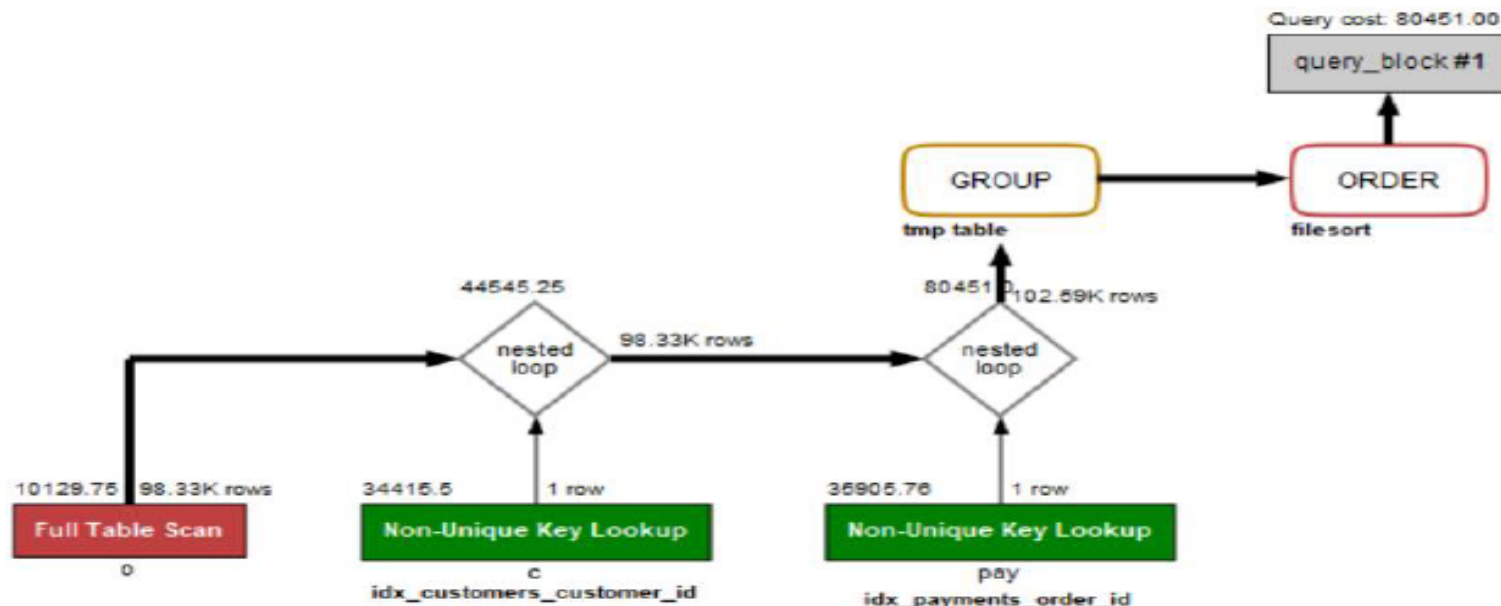
```
select c.customer_id as Customer_id, c.customer_city as Customer_City,
sum(pay.payment_value) as Total_Purchase_Prices
from customers c
join orders o on c.customer_id = o.customer_id
join payments pay on o.order_id = pay.order_id
group by c.customer_id, c.customer_city, o.order_id
order by Total_Purchase_Prices desc;
```

Customer_id	Customer_City	Total_Purchase_Prices
1617b1357756262bfa56ab541c47bc16ec5b2ba62e574342386871631fafd3fc	rio de janeiro	13664.08
c6e2731c5b391845f6800c97401a43a9f48d464a0baaea338cb25f816991ab1f	vila velha	7274.88
3fd6777bbce08a352fdd04e4a7cc8f605455dfa7cd02f13d132aa7a6a9729c6	campo grande	6929.31
df55c14d1476a9a3467f131269c2477fe0a2412720e9ea4f26c1ac985f6a7358	vitoria	6922.21
24bbf5fd2f2e1b359ee7de94defc4a153d979689f636322c62418b6346b1c6d2	marilia	6726.66
1afc82cd60e303ef09b4ef9837c9505cc803a2c412833101651d3f90ca7de24	divinopolis	6081.54
926b6a6fb8b6081e00b335eda5f78d3535a413c7ca3c69756cb75867d6311c0d	araruama	4950.34
e9b0d0eb3015ef1c9ce6cf5b9dbee9f3be2c536886b2ea4668eced3a80dd0bb	golanias	4809.44
eb7a157e8da9c488cd4ddc48711f1097c6695e3b1e48680db36b487419fb0398	maua	4764.34
31e83c01fce824d0ff786fcd48dad009addc91fd9f9c2b3045497b57fc710e820	joao pessoa	4681.78
19b32919fa1198aefc0773ee2e46e69358acc4e2788bf6fc445fddcce9c1db03	sao paulo	4513.32
66657bf1753d82d0a76f2c4719ab8b857d03bf20fa96e80468bbf678eebbcb3f	niteroi	4445.5
39d6658037b1b5a07d0a24d423f0bd19e7c903bf4bb13543e8df947af4f3d9e9	brasilia	4194.76
3c7c62e8d38fb18a33a45db8021f2d690b16ce67087d02bfb33c807e82b1992b	born jesus do ...	4175.26
803cd9b04f9cd252c6a83a2ecd8c22c346bb3c0b1a65c399d0363cefbcc4f37	nova lima	4163.51
a7ea318cb9df2ec79ab37cd7ca2135d	belem	4042.74
	jundiai	4034.44
	sao paulo	4016.91
	rio de janeiro	3979.55
	para de minas	3826.8
	recife	3792.59
	mafra	3782.19
	brasilia	3736.22
	gaspar	3666.42
	brasilia	3602.47
	presidente pr...	3526.46
	joao pessoa	3406.47
	luis eduardo ...	3358.24
	belo horizonte	3351.35
	santa maria	3297.3999999999999
	sao paulo	3256.14

Core Business Insights

1. Which customer has the highest total price of purchases?

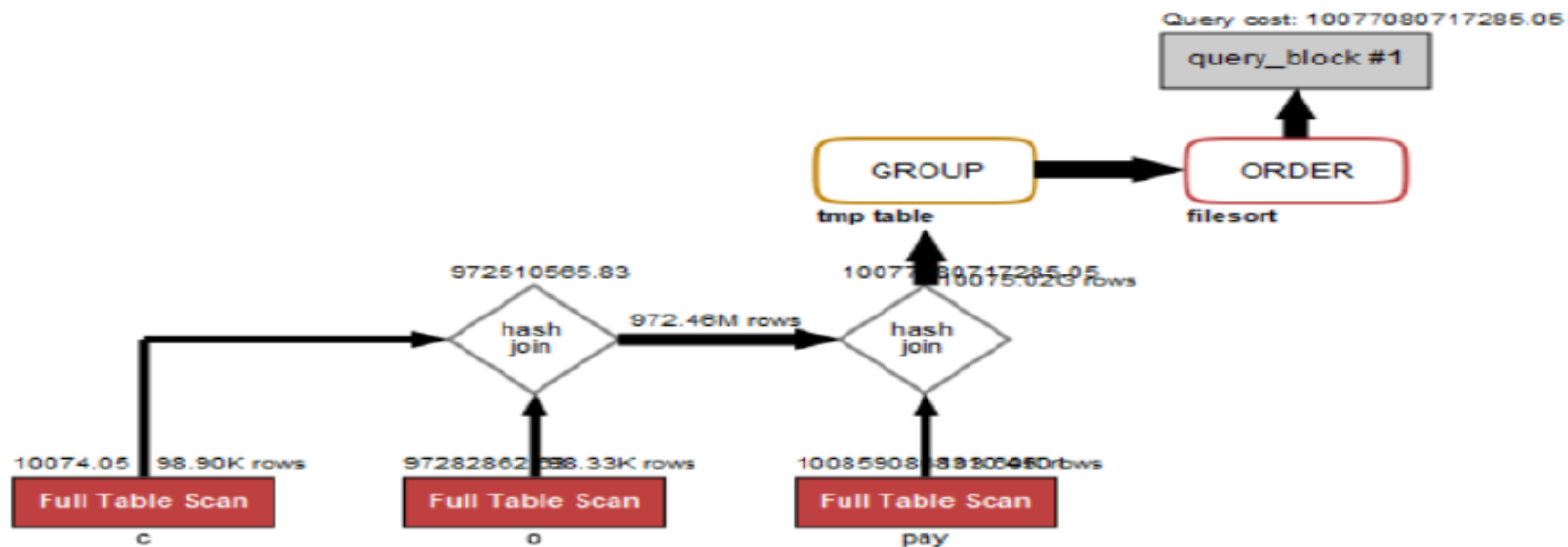
Insight 1 with indexes:



Core Business Insights

1. Which customer has the highest total price of purchases?

Insight 1 without indexes:



Core Business Insights

2. Which product categories have the highest sales?
 - From the result, the top three product categories according to the analysis are "Health & Beauty," "Watches & Gifts," and "Bed & Bath Table."

SQL QUERY 2

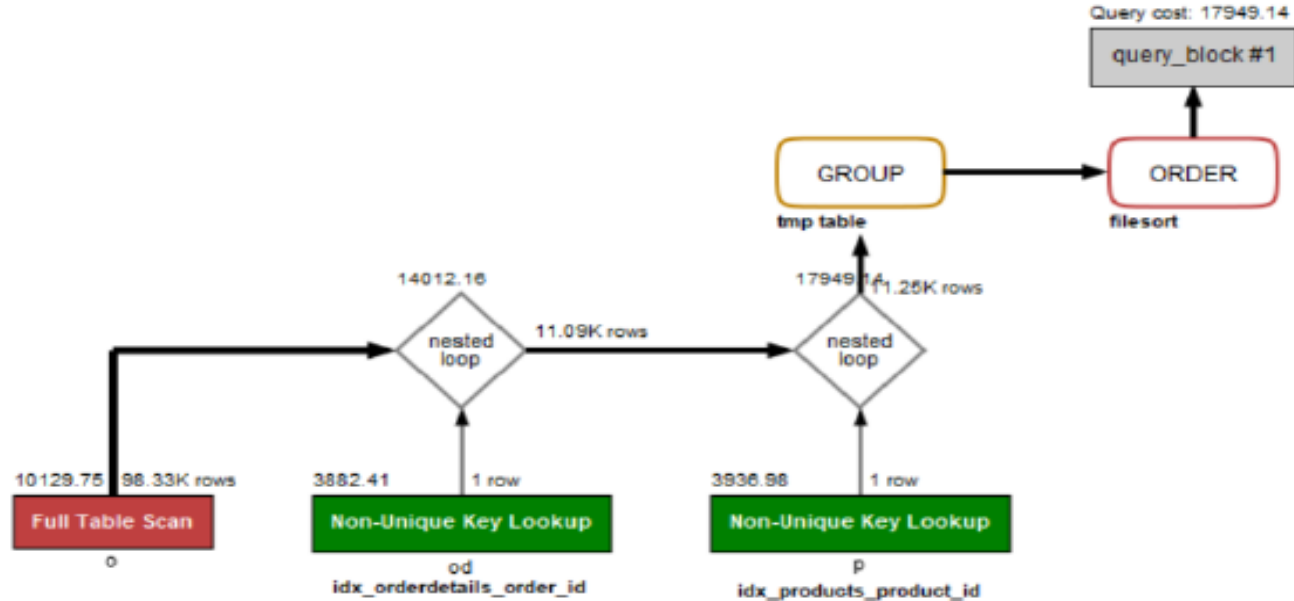
```
select p.product_category_name as Product_Name, sum(od.price + od.freight_value) as  
Total_sales  
from orderdetails od  
join products p on od.product_id = p.product_id  
join orders o on od.order_id = o.order_id  
where o.order_status = "delivered"  
group by p.product_category_name  
order by Total_sales desc;
```

	Product_Name	Total_sales
►	health_beauty	1412089.52999999977
	watches_gifts	1264333.11999999969
	bed_bath_table	1225209.2599999999
	sports_leisure	1118256.9100000008
	computers_accessories	1032723.76999999965
	furniture_decor	880329.9199999993
	housewares	758392.2500000012
	cool_stuff	691680.8900000016
	auto	669454.7499999993
	garden_tools	567145.6799999983
	toys	547061.0599999999
	baby	464766.6499999993
	perfumery	443171.6300000048
	telephony	379202.6200000058
	office_furniture	335211.3600000011
	stationery	269575.0500000005
	pet_shop	250614.1999999996
	computers	228349.7599999998
	musical_instruments	202187.12000000008
	electronics	200723.09000000084
	small_appliances	198108.47000000038
	fashion_bags_accesso...	179888.28999999948
	luggage_accessories	168778.11999999994
	consoles_games	167097.73000000085
	construction_tools_co...	162382.01000000018
	home_appliances_2	118208.52000000006
	home_construction	95022.37000000008
	home_appliances	93301.78
	furniture_living_room	85045.07000000018
	agro_industry_and_co...	76203.29999999997
	home_confort	66438.21999999991

Core Business Insights

2. Which product categories have the highest sales?

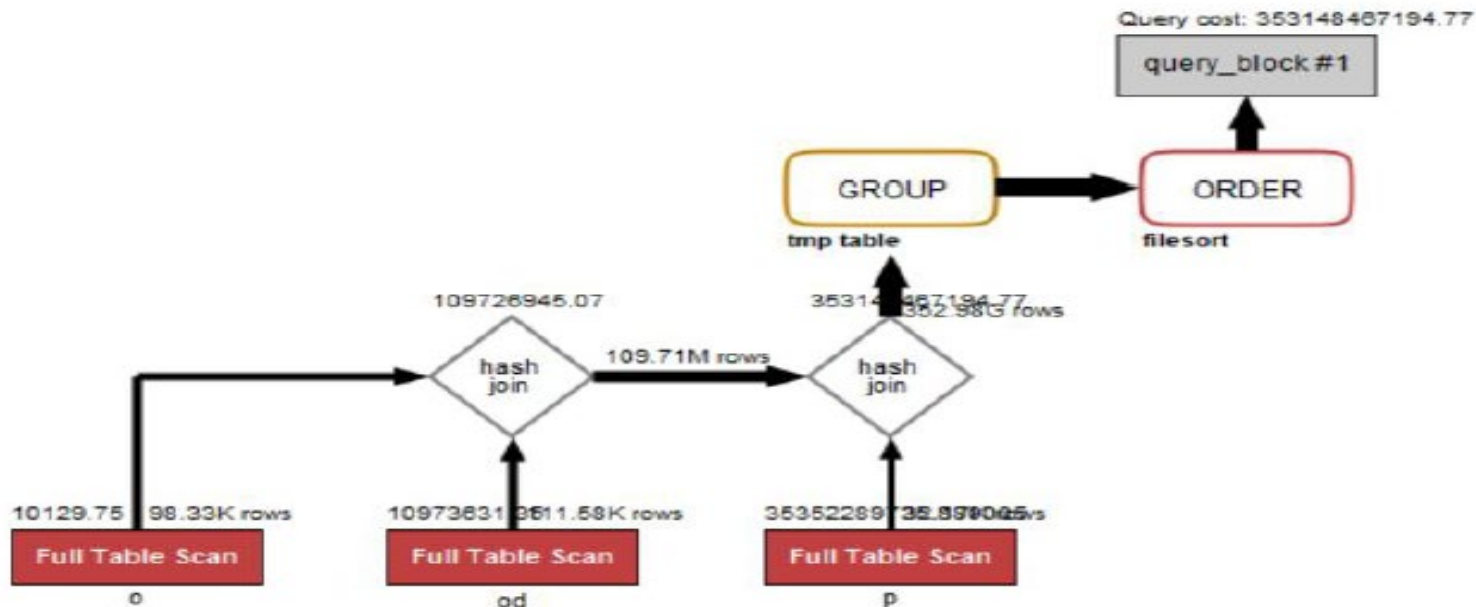
Insight 2 with indexes:



Core Business Insights

2. Which product categories have the highest sales?

Insight 2 without indexes:



Core Business Insights

3. Which products have the highest sale?
- From the result, the products in the "Health & Beauty" category have consistently had strong sales results.

SQL QUERY 3

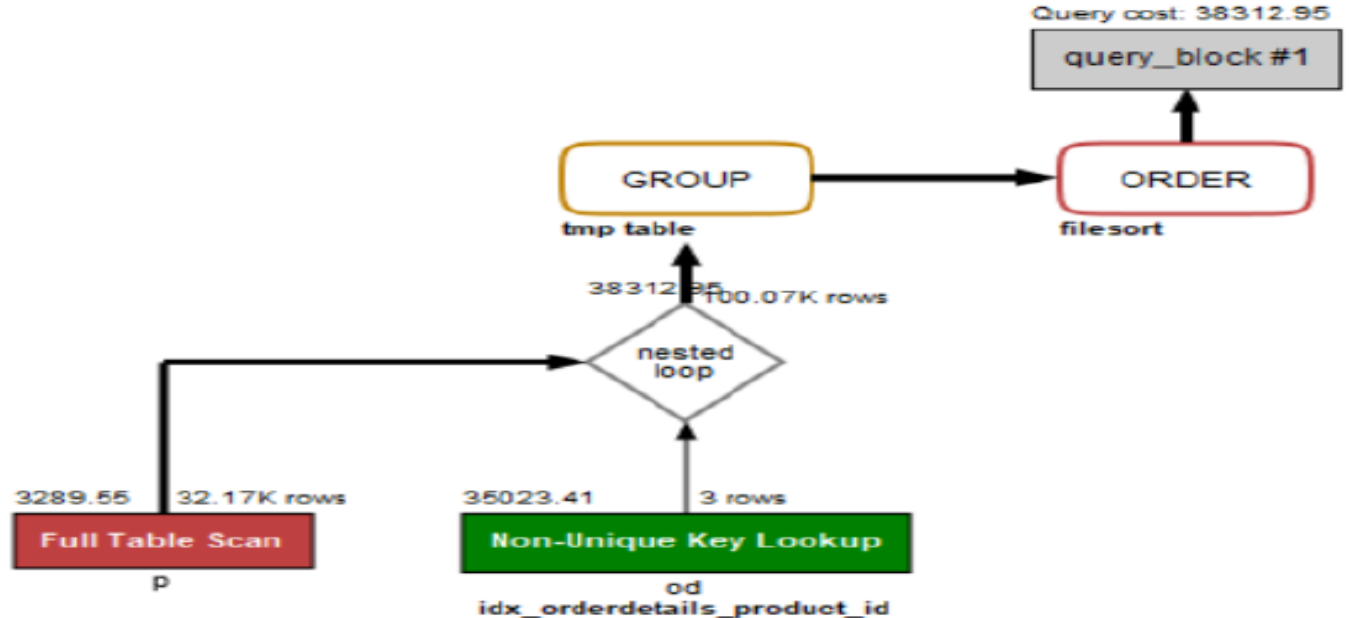
```
select p.product_id as Product_id, p.product_category_name as Product_Name, sum(od.price
+ od.freight_value) as Total_sales
from orderdetails od
join products p on od.product_id = p.product_id
group by p.product_id, Product_Name
order by Total_sales desc;
```

	Product_id	Product_Name	Total_sales
▶	bb50f2e236e5eea0100680137654686c	health_beauty	67606.10000000002
	d1c427060a0f73f6b889a5c7c61f2ac4	computers_accessories	60976.03000000005
	6cdd53843498f92890544667809f1595	health_beauty	59093.98999999994
	99a4788cb24856965c36a24e339b6058	bed_bath_table	51071.599999999989
	d6160fb7873f184099d9bc95e30376af	computers	50326.18
	3dd2a17168ec895c781a9191c1e95ad7	computers_accessories	48212.22000000001
	aca2eb7d00ea1a7b8ebd4e68314663af	furniture_decor	44820.760000000024
	5f504b3a1c75b73d6151be81eb05bdc9	cool_stuff	41725.81000000001
	25c38557cf793876c5abdd5931f922db	baby	40311.94999999999
	53b36df67ebb7c41585e8d54d6772e08	watches_gifts	39957.93000000009
	e0d64dcfaa3b6db5c54ca298ae101d05	watches_gifts	35344.09000000003
	422879e10f46682990de24d770e7f83d	garden_tools	34201.26000000001
	d285360f29ac7fd97640bf0baef03de0	watches_gifts	33607.57000000001
	f1c7f353075ce59d8a6f3cf58f419c9c	bed_bath_table	33316.76000000002
	7a10781637204d8d10485c71a6108a2e	watches_gifts	32790.73999999998
	f819f0c84a64f02d3a5606ca95edd272	watches_gifts	30449.690000000006
	588531f8ec37e7d5ff5b7b22ea0488f8	computers	29080.209999999985
	389d119b48cf3043d311335e499d9c6b	garden_tools	28682.679999999996
	a62e25e09e05e6faf31d90c6ec1aa3d1	watches_gifts	27998.30000000005
	368c6c730842d78016ad823897a372db	garden_tools	27984.4
	53759a2eccddad2bb87a079a1f1519f73	garden_tools	27268.22999999998
	2b4609f8948be18874494203496bc318	health_beauty	26659.740000000005
	16c4e87b98a9370a9cbc3a4658a3f45b	musical_instruments	26630.149999999994
	52c80cedd4e90108bf4fa6a206ef6b03	garden_tools	25886.25999999999
	19c91ef95d509ea33eda93495c4d3481	health_beauty	24326.250000000004
	461f43be3bdf8844e65b62d9ac2c7a5a	watches_gifts	23215.830000000005
	a5215a7a9f46c4185b12f38e9ddf2abc	computers	22411.020000000004
	fd0065af7f09af4b82a0ca8f3eed1852	auto	22367.88
	1dec4c88c685d5a07bf01dc0f8bf9f8	auto	22228.510000000002
	bc4cd4da98dd128c39bf0b8c2674032f	computers	22184.439999999995
	c4baedd846ed09b85f78a781b522f126	garden_tools	20950.120000000006

Core Business Insights

3. Which products have the highest sale?

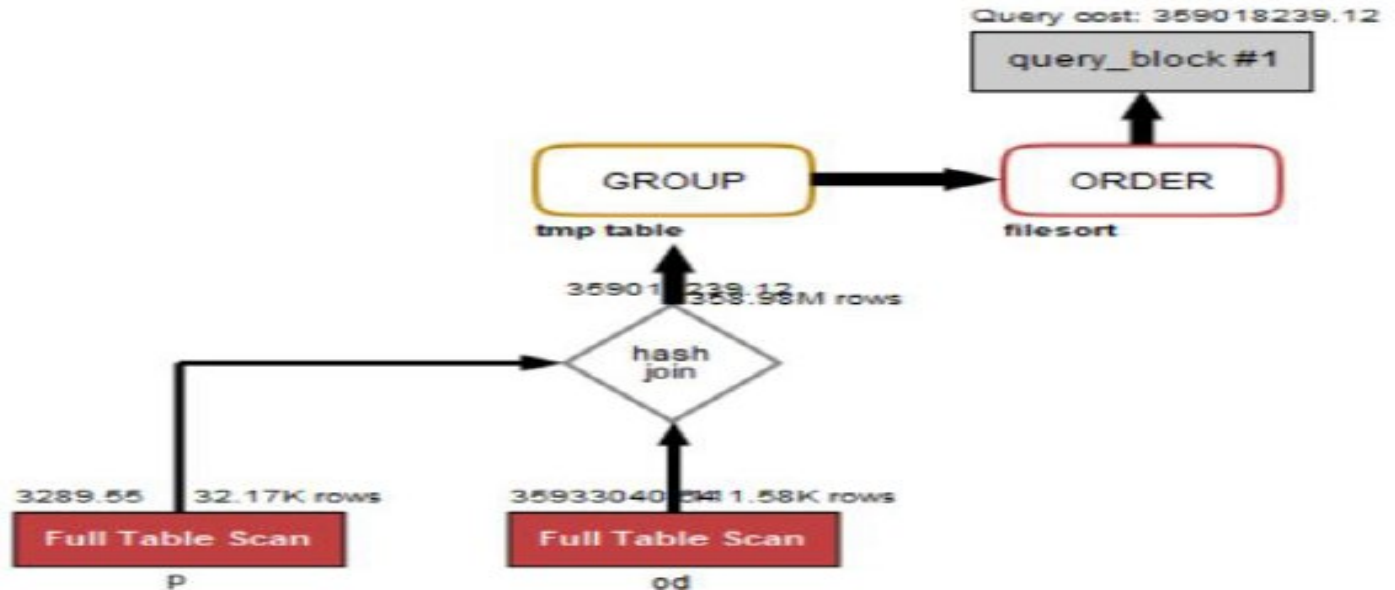
Insight 3 with indexes:



Core Business Insights

3. Which products have the highest sale?

Insight 3 without indexes:





Core Business Insights

4. Which payment methods are most commonly used?
- From the result, credit card payments are the most popular with 76795 transactions and highlight the need for safe and practical credit card options.

SQL QUERY 4

```
select distinct(payment_type) as Payment_Types, count(payment_type) as  
Times_Of_Payments  
from payments  
group by payment_type;
```

	Payment_Types	Times_Of_Payments
▶	credit_card	76795
	boleto	19784
	voucher	5775
	debit_card	1529
	not_defined	3

Core Business Insights

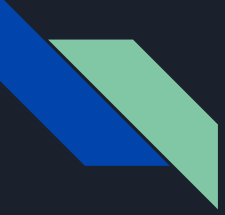
5. Who are the best performing sellers?

- From the result, respective seller IDs (4869f7a5dfa277a7dca64, 53243585a1d6dc264302, and 4a3ca9315b744ce9f8e9) which the top three sellers have impressive total sales

SQL QUERY 6

```
SELECT
    s.seller_id,
    COUNT(od.order_item_id) AS order_count,
    SUM(od.price) AS total_revenue
FROM OrderDetails od
JOIN Sellers s ON od.seller_id = s.seller_id
GROUP BY s.seller_id
ORDER BY total_revenue DESC;
```

	seller_id	order_count	total_revenue
▶	4869f7a5dfa277a7dca6462dcf3b52b2	1156	229472.62999999981
	53243585a1d6dc2643021fd1853d8905	410	222776.049999999952
	4a3ca9315b744ce9f8e9374361493884	1987	200472.91999999949
	fa1c13f2614d7b5c4749cbc52fecda94	586	194042.029999999846
	7c67e1448b00f6e969d365cea6b010ab	1364	187923.88999999995
	7e93a43ef30c4f03f38b393420bc753a	340	176431.869999999982
	da8622b14eb17ae2831f4ac5b9dab84a	1551	160236.569999999538
	7a67c85e85bb2ce8582c35f2203ad736	1171	141745.530000000177
	1025f0e2d44d7041d6cf58b6550e0bfa	1428	138968.54999999995
	955fee9216a65b617aa5c0531780ce60	1499	135171.699999999914



Thank you!

